

Kernel Triviality for Every Admissible Heterogeneous Horizon

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Source question

Kreisbeck and Schönberger introduce heterogeneous nonlocal gradients whose interaction horizon $\delta(x)$ stays inside the domain and vanishes smoothly at its boundary [1]. Their Theorem 4.17 proves, with no amplitude restriction, that the nullspace is finite-dimensional and consists of smooth functions. When the maximal horizon is sufficiently small (“mild variation”), they prove that the nullspace consists exactly of constants. Remark 4.18 asks whether the same conclusion holds without that smallness assumption.

Remark 4.18. While our current proof does not extend to this case, we expect that (4.14) holds also when δ is not mildly varying. In fact, this is corroborated by some numerical simulations, which show that the eigenvalues of the discretization of $\mathcal{Q}_{\rho(\cdot)}^{\Omega}$ are all positive and bounded away from zero. $\triangle$

The theorem below gives an affirmative answer in every space dimension.

Setting

Let $\Omega \subset \mathbb{R}^n$ be a bounded connected Lipschitz domain and let $p \in (1, \infty)$. The horizon $\delta \in C^\infty(\mathbb{R}^n)$ satisfies the source paper’s assumptions; in particular,

$$\delta > 0 \text{ in } \Omega, \quad \delta = 0 \text{ on } \Omega^c, \quad \delta(x) \leq \text{dist}(x, \Omega^c).$$

Let $\bar{\rho}_1$ be the radial representative of the source kernel and put

$$Q(r) = Q_{\rho_1}(r) = \int_r^1 \frac{\bar{\rho}_1(s)}{s} ds, \quad g(r) = r^{n-2} \bar{\rho}_1(r).$$

Hypothesis H1 says that g is decreasing. The averaging operator on $\mathbb{R}^n$ is

$$(\mathcal{Q}_{\rho(\cdot)} f)(x) = \begin{cases} \int_{\mathbb{R}^n} \delta(x)^{-n} Q\left(\frac{|x-y|}{\delta(x)}\right) f(y) dy, & x \in \Omega, \\ f(x), & x \in \Omega^c. \end{cases}$$

The source normalization gives $\int_{\mathbb{R}^n} Q(|z|) dz = 1$, and

$$\mathcal{D}_{\rho(\cdot)} u = \mathcal{Q}_{\rho(\cdot)}^{\Omega} \nabla u.$$

Idea of the proof

The source proof already turns a kernel element u into a smooth, compactly supported vector field $\tilde{w}$ with $\mathcal{Q}_{\rho(\cdot)}\tilde{w} = 0$ and $\tilde{w}|_{\Omega} = \nabla u$. The remaining task is therefore a scalar injectivity statement for the components of $\tilde{w}$.

For $n \geq 2$, solve $\Delta F = \tilde{w}_j$ by a Newtonian potential. Two radial integrations by parts show that $\mathcal{Q}_{\rho(\cdot)}\Delta F$ is a nonnegative mixture of

$$\text{spherical mean of } F \text{ about } x - F(x).$$

This gives a maximum principle. The Newtonian potential decays at infinity, so it must vanish. In dimension two, decay requires zero total mass; that mass vanishes because $\tilde{w} = \nabla u$ in Ω , while u is constant on every boundary curve. Dimension one is handled by the corresponding centered-second-difference maximum principle.

The spherical-mean identity

Lemma 1. *Let $F \in C^2(\mathbb{R}^n)$, $x \in \Omega$, and $d = \delta(x) > 0$. Define*

$$M_x(r) = \frac{1}{|\mathbb{S}^{n-1}|} \int_{\mathbb{S}^{n-1}} F(x + dr\theta) d\theta, \quad 0 \leq r \leq 1.$$

Then

$$(\mathcal{Q}_{\rho(\cdot)}\Delta F)(x) = \frac{|\mathbb{S}^{n-1}|}{d^2} \int_0^1 (-g'(r))(M_x(r) - M_x(0)) dr. \quad (1)$$

The integral is understood improperly at zero.

Proof. The standard spherical-mean identity gives

$$M_x''(r) + \frac{n-1}{r} M_x'(r) = \frac{d^2}{|\mathbb{S}^{n-1}|} \int_{\mathbb{S}^{n-1}} \Delta F(x + dr\theta) d\theta.$$

Using polar coordinates and then integrating by parts,

$$\begin{aligned} (\mathcal{Q}_{\rho(\cdot)}\Delta F)(x) &= \frac{|\mathbb{S}^{n-1}|}{d^2} \int_0^1 Q(r)(r^{n-1}M_x'(r))' dr \\ &= \frac{|\mathbb{S}^{n-1}|}{d^2} \int_0^1 r^{n-2} \bar{\rho}_1(r) M_x'(r) dr \\ &= \frac{|\mathbb{S}^{n-1}|}{d^2} \int_0^1 (-g'(r))(M_x(r) - M_x(0)) dr. \end{aligned}$$

Here $Q' = -\bar{\rho}_1/r$.

We verify the singular boundary terms. At $r = 1$, both Q and g vanish because ρ_1 has compact support in the open unit ball. At zero, $M_x'(r) = O(r)$ and $M_x(r) - M_x(0) = O(r^2)$. Moreover,

$$r^n Q(r) = \int_r^1 \left(\frac{r}{s}\right)^n s^{n-1} \bar{\rho}_1(s) ds \longrightarrow 0$$

by dominated convergence and radial integrability. Finally, monotonicity of g gives

$$\int_{r/2}^r s^{n-1} \bar{\rho}_1(s) ds = \int_{r/2}^r s g(s) ds \geq \frac{3}{8} r^2 g(r) = \frac{3}{8} r^n \bar{\rho}_1(r).$$

The left side tends to zero. Thus $Q(r)r^{n-1}M_x'(r) \rightarrow 0$ and $g(r)(M_x(r) - M_x(0)) \rightarrow 0$, completing both integrations by parts. $\square$

A maximum principle

Define

$$\mathcal{L}F := \mathcal{Q}_{\rho(\cdot)}\Delta F.$$

Lemma 2. *Let $F \in C^2(\mathbb{R}^n)$ satisfy $\mathcal{L}F = 0$. If $\bar{\Omega} \subset B_R$, then*

$$\inf_{\partial B_R} F \leq F(x) \leq \sup_{\partial B_R} F \quad (x \in B_R).$$

Proof. Since g is decreasing, $-g' \geq 0$. Lemma 1 therefore implies that $\mathcal{L}G(x_0) \leq 0$ whenever $x_0 \in \Omega$ is a maximum of G over B_R : the interaction ball is contained in $\Omega \subset B_R$, and every spherical mean is at most $G(x_0)$. If $x_0 \in \Omega^c$, then $\mathcal{L}G(x_0) = \Delta G(x_0) \leq 0$ at an interior maximum.

For $\varepsilon > 0$, put $G_\varepsilon = F + \varepsilon|x|^2$. The normalization of Q gives

$$\mathcal{L}G_\varepsilon = 2n\varepsilon > 0.$$

Consequently G_ε cannot attain its maximum in the interior of B_R . Hence, for $x \in B_R$,

$$F(x) + \varepsilon|x|^2 \leq \sup_{\partial B_R} F + \varepsilon R^2.$$

Letting $\varepsilon \downarrow 0$ proves the upper bound. Applying the same argument to $F - \varepsilon|x|^2$ and interior minima proves the lower bound. $\square$

Lemma 3. *Let $f \in C_c^\infty(\mathbb{R}^n)$ and suppose $\mathcal{Q}_{\rho(\cdot)}f = 0$.*

1. *If $n \geq 3$, then $f = 0$.*
2. *If $n = 2$ and $\int_{\mathbb{R}^2} f = 0$, then $f = 0$.*

Proof. Let $F = \Gamma_n * f$, where Γ_n is the Newtonian kernel normalized so that $\Delta \Gamma_n = \delta_0$; see, for example, [2, Section 2.2]. Then $F \in C^\infty(\mathbb{R}^n)$, $\Delta F = f$, and $\mathcal{L}F = \mathcal{Q}_{\rho(\cdot)}f = 0$.

For $n \geq 3$, compact support of f gives $F(x) = O(|x|^{2-n})$, so $F(x) \rightarrow 0$ at infinity. For $n = 2$,

$$F(x) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \log|x-y| f(y) dy.$$

The coefficient of $\log|x|$ is $(2\pi)^{-1} \int f$, hence the stated zero-mass assumption gives $F(x) = O(|x|^{-1})$. In either case, $F \rightarrow 0$ uniformly on large spheres.

Apply Lemma 2 on B_R and let $R \rightarrow \infty$. It follows that $F = 0$, and therefore $f = \Delta F = 0$. $\square$

The two-dimensional moment

Lemma 4. *Suppose $n = 2$, $u \in C^\infty(\bar{\Omega})$, and $w \in C_0^\infty(\bar{\Omega}; \mathbb{R}^2)$ satisfies $w = \nabla u$ in Ω . Then*

$$\int_{\mathbb{R}^2} w_j dx = 0, \quad j = 1, 2.$$

Proof. Because the zero extension of w is smooth, $w = 0$ on $\partial\Omega$. Hence $\nabla u = 0$ there. The boundary of a bounded planar Lipschitz domain is a finite union of closed Lipschitz curves Γ_ℓ , and u is constant, say c_ℓ , on each curve. The divergence theorem gives

$$\int_{\mathbb{R}^2} w_j dx = \int_{\Omega} \partial_j u dx = \sum_{\ell} c_\ell \int_{\Gamma_\ell} \nu_j d\mathcal{H}^1.$$

The integral of the normal vector around every closed Lipschitz curve is zero (equivalently, apply the divergence theorem to a constant vector field on the region bounded by that curve). Every summand vanishes. $\square$

Dimension one

Lemma 5. *In the source setting with $n = 1$, $\mathcal{D}_{\rho(\cdot)}u = 0$ implies that u is constant, for every admissible horizon.*

Proof. The no-smallness part of the source theorem gives $u \in C^\infty([a, b])$, where $\Omega = (a, b)$. Choose F with $F'' = u'$, put

$$h(r) = \frac{\bar{\rho}_1(r)}{r},$$

and note that H1 makes h decreasing and $-h' > 0$ near zero. The identity $\mathcal{D}_{\rho(\cdot)}u = \mathcal{Q}_{\rho(\cdot)}^\Omega u'$, followed by two one-dimensional integrations by parts, gives

$$0 = \frac{1}{\delta(x)^2} \int_0^1 (-h'(t))(F(x + \delta(x)t) + F(x - \delta(x)t) - 2F(x)) dt.$$

Indeed, H1 gives an $\iota > 0$ such that $t^\iota h(t)$ is decreasing near zero, while H0 gives $h(t) > 0$ there; hence

$$-h'(t) \geq \frac{\iota}{t} h(t) > 0.$$

The endpoint terms vanish by the same estimates used in Lemma 1.

Subtract from F its secant line on $[a, b]$, obtaining G with $G(a) = G(b) = 0$. At a positive interior maximum every centered second difference is nonpositive. Since $-h' > 0$ near zero, equality forces G to equal its maximum in a neighborhood. Thus the maximum set is nonempty, open, and closed in (a, b) , contradicting the boundary values. The same argument excludes a negative minimum. Hence $G = 0$, F is affine, and $u' = F'' = 0$. $\square$

Full solution

Theorem 1. *Under all the hypotheses of Kreisbeck and Schönberger [1, Theorem 4.17], but with no mild-variation or small-amplitude assumption on the admissible horizon, one has*

$$\ker \mathcal{D}_{\rho(\cdot)} = \{u : \Omega \rightarrow \mathbb{R} : u \text{ is constant}\}.$$

This holds for every $n \geq 1$ and every $p \in (1, \infty)$.

Proof. Constants plainly lie in the kernel. Let $u \in \ker \mathcal{D}_{\rho(\cdot)}$. The no-smallness part of the source proof constructs

$$\tilde{w} = \nabla \tilde{u} - \mathcal{D}_{\rho(\cdot)}^{\mathbb{R}^n} \tilde{u} \in C_0^\infty(\bar{\Omega}; \mathbb{R}^n)$$

such that

$$\mathcal{Q}_{\rho(\cdot)} \tilde{w} = 0, \quad \tilde{w}|_\Omega = \nabla u.$$

These are equations (4.15)–(4.16) and the surrounding argument in [1, Theorem 4.17]; neither uses mild variation.

If $n = 1$, Lemma 5 gives the conclusion directly. Suppose $n \geq 2$. Apply Lemma 3 to each component $\tilde{w}_j$. For $n \geq 3$, no additional condition is needed. For $n = 2$, Lemma 4 supplies $\int \tilde{w}_j = 0$. Thus $\tilde{w} = 0$ in every case. Consequently $\nabla u = \tilde{w}|_\Omega = 0$, and connectedness of Ω makes u constant. $\square$

Verification notes

The script `code/verify_spherical_mean_identity.py` checks Lemma 1 exactly in dimension three for $F(y) = |y|^4$, an arbitrary center, arbitrary scale, and a nontrivial radial profile. It returns `symbolic_difference=0` and `PASS`. The script `code/verify_second_difference_identity.py` performs the corresponding exact one-dimensional check and also passes. They were run with

```
conda run -no-capture-output -n sandbox python code/<script>.py.
```

The two spectral scripts are exploratory only. They found no nonpositive eigenvalue over the admissible amplitude range and thousands of irregular horizons, but no computational observation is used in the proof.

Novelty check and limitations

The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2509.03468, the exact title, heterogeneous horizons, kernel triviality, variable-bandwidth injectivity, Newtonian potentials, and spherical means. The only indexed result was the dimension-one precursor now superseded by this packet.

A bounded external search on July 23, 2026 used arXiv and exact combinations of “heterogeneous horizon”, “nonlocal gradient kernel”, “mildly varying”, “Newtonian potential”, and “spherical mean maximum principle”. It found the source preprint and unrelated nonlocal/variable-bandwidth work, but no answer to Remark 4.18. The official arXiv record still lists only version 1, submitted September 3, 2025. Novelty is plausible, not certified by an exhaustive review.

The proof resolves precisely the source expectation under the source hypotheses. It does not claim injectivity of $\mathcal{Q}_{\rho(\cdot)}$ on every scalar L^p function in dimension two: Lemma 3 uses zero mass there. The source auxiliary gradient has exactly that cancellation, which is all Theorem 1 requires.

Human review recommendation

Review as a claimed full solution. The main checks are the singular endpoint limits in Lemma 1, the strict quadratic perturbation in Lemma 2, and the dimension-two boundary-component calculation in Lemma 4. A reviewer should also confirm the cited source proof indeed supplies $\tilde{w} \in C_0^\infty(\overline{\Omega})$ before mild variation is invoked.

References

- [1] C. Kreisbeck and H. Schönberger, *Local boundary conditions in nonlocal hyperelasticity via heterogeneous horizons*, arXiv:2509.03468v1, 2025. <https://arxiv.org/abs/2509.03468>.
- [2] L.C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics 19, American Mathematical Society, 2010.